

Motor Imagery Multi-classification based on Canonical Correlation Analysis Feature Fusion

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Abstract: The application of discrete wavelet transform (DWT) or common space transform (CSP) extracts the features of motor imaginary (MI) electroencephalogram (EEG) signals. However, these feature extraction algorithms suffer from some drawbacks, such as the low generalization ability of CSP dealing with different subjects and the dependence of time-frequency window of DWT. In this paper, canonical correlation analysis (CCA) is proposed to fuse two features extracted by DWT and CSP so as to eliminate the noise with low correlation. CCA retains not only the advantage of CSP requiring less data in the training set, but also the advantage of DWT having high adaptability to multi-classification problems and nonlinear signals. The experimental results on BCI2005desc-IIIa and BCI2008desc-IIa EEG dataset show that the average accuracy of proposed multi-domain feature fusion model based on CCA is more than 87.50% for four-class MI tasks on BCI2005desc-IIIa data set, which is higher than non-feature-fusion methods, and is 78.07% for four-class MI tasks on BCI2008desc-IIa data set, which is higher than other feature extraction methods.

Key Words: Canonical Correlation Analysis (CCA), Common spatial Patterns (CSP), Discrete Wavelet Transform (DWT), Brain-Computer Interface (BCI), Feature Fusion

1 Introduction

With the rapid development of biotechnology and computer science, people are exploring approaches to use signals from brain activity rather than peripheral muscles to provide communication and control ability, known as Brain-Computer Interface (BCI). With the further development of BCI, the application of this communication mode has attracted people's attention from both civil and military field. Motor Imagery (MI) is a mental process when people imagine a certain type of physical activity, which can be detected by a special electroencephalogram (EEG) monitoring device. MI, a signal trigger type of BCI, has been proved to be effective in neuron-rehabilitation, functional motor disorder rehabilitation and central nervous system reorganization[1].

The feature extraction methods of EEG signals are based on traditional statistical methods, such as Common Spatial Patterns (CSP), Discrete Wavelet Transform (DWT) and Hilbert transform, etc.[2] Canonical Correlation Analysis (CCA), a feature statistical method, was used to process Event-Related Synchronization[3] and Desynchronization signals for MI signals classification in the early stage[4]. With the development of machine learning and deep learning, many classification methods like Support Vector Machine (SVM)[5] and neural network[6] have been used as classifiers in many MI classification studies.

However, the above methods all only use a single domain (time domain, frequency domain or spatial domain) with other domain invalid, which leads to problems like being sensitive to interference, long processing time, being greatly affected by signal non-linearity, and large number of samples in training set[7].

Aiming at the above problems, the multi-domain feature fusion model based on CCA is proposed in this paper.

Firstly, DWT and CSP are used to extract the features of EEG signals in the time-frequency domain and spatial domain respectively. DWT extracts different frequency components of non-stationary signals and represent the differences of signals in time domain with wavelet coefficients. CSP extracts the synchronous features of multi-channel EEG signals and shows the spatial differences of the signals. Then CCA is used to perform feature fusion on the features extracted by the two above methods, solving the shortcoming of one single domain. Finally, the MKL-SVM optimized by particle swarm optimization and simulated annealing (PSO-SA) is used to classify the fused feature. The proposed method reaches classification accuracy of more than 87.50% for four-class MI data set in BCI Composition 2005 and 78.07% for four-class MI data set in BCI Composition 2008.

The rest of this paper is organized as follows. The feature extraction algorithms of DWT and CSP are reviewed in Section 2. The proposed multi-domain feature fusion model based on CCA is illustrated in Section 3. Experimental data and result analysis are represented in Section 4. The paper is finally concluded in Section 5.

2 Feature Extraction

Feature extraction of the EEG signals refers to extract the advanced signal features which reflect the user's intention after preprocessing and digital processing[8].

2.1 Discrete Wavelet Transform

The time-frequency window size of DWT is related to the frequency of the signal. The essence of DWT is to decompose the original signal into a series of wavelet functions. With the quality of time-frequency localization and multi-resolution, DWT is used to extract the time-frequency domain features of MI EEG signals, frequency decomposition at low frequencies and time decomposition at high frequencies. The multi-scale analysis is suitable for feature extraction of non-stationary EEG signals.

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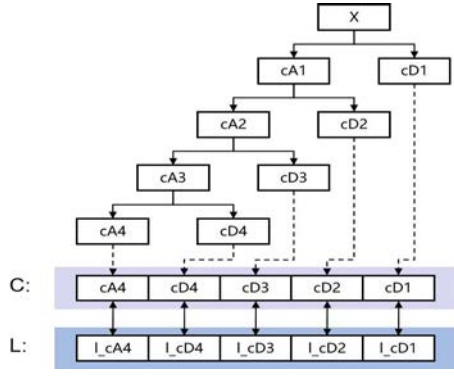


Fig. 1: Decomposition structure of 4-layer-DWT

Assuming that the discrete signals collected by the EEG acquisition equipment are $x(n)$, the collected signals is decomposed by DWT in preset finite layer. Assuming the output of layer i are detail component D_i and approximate component A_i , A_i is further decomposed according to the above process at the $i+1$ layer. This process is repeated until a preset number of layers is reached, and the detailed process of 4-layer DWT is shown in Fig. 1. So, DWT can represent as[9]:

$$x(n) = cA_L + \sum_{i=1}^L cD_i \quad (1)$$

Table 1: Wavelet decomposition corresponds to the frequency band of each layer

Signal Component	Layer Number	Frequency Band / Hz
cD1	1	62.5 ~ 125
cD2	2	31.25 ~ 62.5
cD3	3	15.625 ~ 31.25
cD4	4	7.8125 ~ 15.625
cA4	4	0 ~ 7.8125

Research on MI EEG signals has shown that when people imagine unilateral limb movements, ERD potentials are produced by α and β rhythm in the frequency band between 8 and 30 Hz[10]. The recognition rate using db4 wavelet functions is more ideal in the study[11]. In this paper, db4 wavelet is selected to decompose MI EEG signals in four layers, the specific results are shown in Table 1. And the detailed components of cD3 and cD4, which are basically coincident with the frequency bands of α and β rhythms, were selected for analysis.

After extracting the features using DWT, the time-frequency features of each EEG sample with $I \times 8$ dimensions were obtained by taking the averages and energy averages of cD3 and cD4. Then the normalization processing is carried out in order to unify the statistical distribution law of the extracted feature vectors.

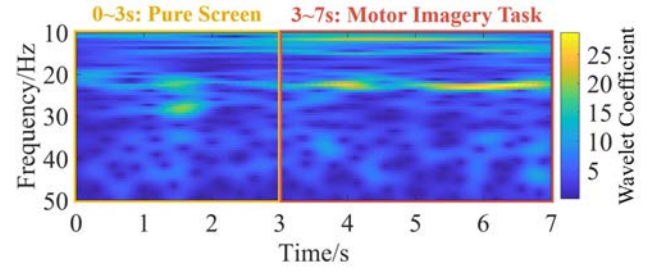


Fig. 2: Experimental results of wavelet transform

The amplitude of wavelet coefficient corresponds to the shade of color Fig. 2 shows that the wavelet coefficients at different time frequencies are significantly different, where the wavelet coefficients of α and β rhythm vary from the phase of pure screen to the phase during the MI tasks.

2.2 Common Spatial Patterns

CSP is one of the most popular algorithms for feature extraction using spatial filtering and classification in two class of MI signals. As for multi-classification tasks, it is necessary to introduce one-versus-rest CSP to extend the two-class to multi-class feature extraction. The principle is to take one of the multi-class signals as the first class and the other signals as the second class for OVR-CSP. Feature extraction is carried out for each class respectively. For N-class EEG signals, there are finally N 2-class CSP filters.

The process of calculating the 2-class CSP filters is as follows:

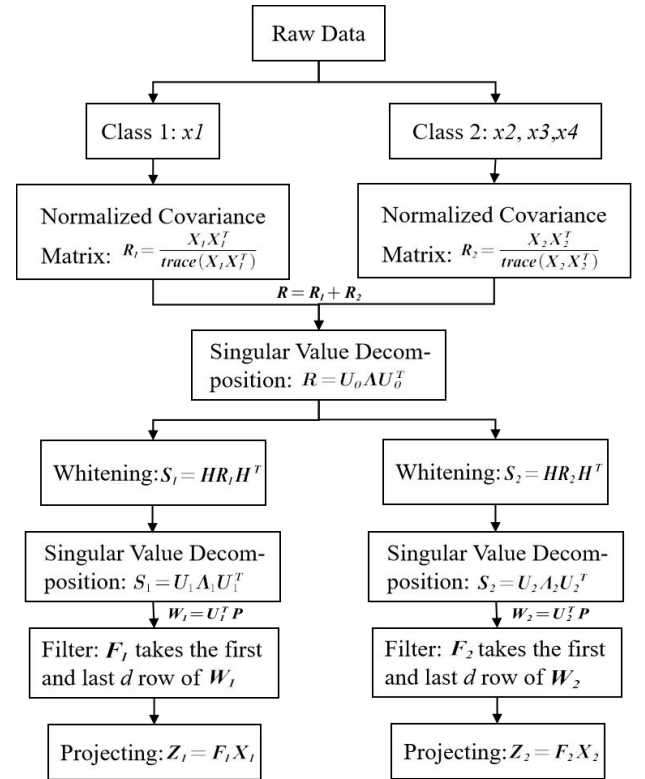


Fig. 3: The Calculative Process of 2-class CSP Filter

For each 2-class CSP filter, we obtain $I \times 2d$ dimensional feature in spatial domain, where d is the number of vectors selected for the projection matrix.

3 Multi-domain Feature Fusion Model

Fig.4 shows the multi-domain feature fusion model framework proposed in this work. The pattern recognition system of MI EEG signals includes the following 2 parts, feature extraction and classification. Feature extraction is mainly about reducing the dimensions of EEG signals and extract the feature for classification.

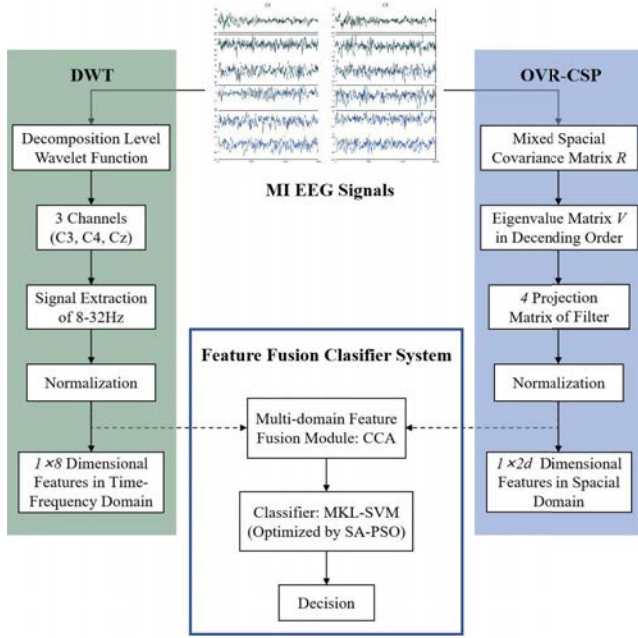


Fig. 4: Block Diagram of proposed Multi-domain Feature Fusion Model based on CCA

The feature extraction algorithm used in this work is DWT and OVR-CSP. The time-frequency domain and spatial domain characteristics of MI signal can be extracted respectively. After feature fusion by CCA, MKL-SVM is used as the classifier in this paper, and PSO-SA algorithm is used to adjust and optimize this classifier.

Canonical Correlation Analysis (CCA), a typical unsupervised learning algorithm, is used to fuse two sets of feature vectors extracted from spatial domain with CSP and time-frequency domain with DWT, retaining discriminant information and removing redundant information in the feature sets. CCA focuses on the mutual information between the two feature vectors, aiming to maximize the variable correlation between the two sets of feature vectors.

Assuming that the feature sets extracted by CSP and DWT is $X = [X_1, X_2, \dots, X_n]$, $Y = [Y_1, Y_2, \dots, Y_n]$, where

$X \in R^{p \times n}$, $Y \in R^{q \times n}$, and n is the number of samples. The feature fusion vectors are standardized to obtain the feature sets X and Y with zero-mean and one-variance. CCA finds the projection vector a and b for each variable to maximize the variables correlated in the projected space in the two feature sets.

When $S_{XY} = \text{cov}(X, Y)$, the optimization objective is transformed into:

$$\arg \max_{a,b} \frac{a^T S_{XY} b}{\sqrt{a^T S_{XX} a} \sqrt{b^T S_{YY} b}} \quad (2)$$

For the above optimization objectives, singular value decomposition (SVD) is used to solve the function optimization easily. First, the matrix is standardized as follows:

$$\arg \max_{a,b} u^T S_{XX}^{-\frac{1}{2}} S_{XY} S_{YY}^{-\frac{1}{2}} v \quad (3)$$

$$s.t. \quad u^T u = 1, v^T v = 1$$

Where u and v are regarded as the left and right singular vectors corresponding to a singular value of the matrix:

$$M = u^T S_{XX}^{-\frac{1}{2}} S_{XY} S_{YY}^{-\frac{1}{2}} v \quad (4)$$

Using SVD, $M = U \Sigma V^T$ is decomposed, where U and V are composed of left and right singular vectors of M are respectively compose matrix U and V , and Σ is the diagonal matrix composed of singular value of M . Since all the columns of U and V are orthonormal bases, $u^T U$ and $V^T v$ obtain a vector with only one scalar quantity of 1 and the rest of 0. Then the function is:

$$u^T S_{XX}^{-\frac{1}{2}} S_{XY} S_{YY}^{-\frac{1}{2}} v = u^T U \Sigma V^T v = \sigma_{uv} \quad (5)$$

The maximum singular value of M is also the maximum of the optimization target, which is the maximum correlation coefficient between X and Y . Therefore, the linear coefficients of a and b are obtained by using the corresponding left and right singular vectors:

$$\begin{cases} a = S_{XX}^{-\frac{1}{2}} u \\ b = S_{YY}^{-\frac{1}{2}} v \end{cases} \quad (6)$$

Thus, the multi-domain fusion feature vector Z is obtained by combining the spatial feature vector X extracted by CSP and the time-frequency feature vector Y extracted by DWT using CCA[12]:

$$Z = a^T X_i + b^T Y_i = \begin{bmatrix} a^T & b^T \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \end{bmatrix} \quad (7)$$

During the training phase, the multi-domain fused feature vector Z is stored as the template in the gallery. While in the testing phase, the multi-domain fused feature vector of test sample can be classified by SVM classifier.

4 Experiments and Results

4.1 Database Description

In this paper, we used the BCI2005desc-IIIa[13] and BCI2008desc-IIa[14] EEG dataset as experimental data, which had been widely used in the study of MI for comparative analysis. These two datasets consist of four different MI tasks, including imaging to move left hands, right hands, feet, and tongue whose labels are respectively divided into class 1, class 2, class 3, and class 4.

MI data set of BCI2005desc-IIIa recorded from a 64-channel EEG amplifier with a 250-Hz sampling rate and band-pass filtered between 1 and 50 Hz. The timing scheme consists of a blank screen then a cue for 3 seconds followed by a period of MI task of 4s, which gives an arrow for 1s. The data has 240 trials (60 trials per class) for training and evaluation data.

MI data set of BCI2008desc-IIa recorded from 22 Ag/AgCl electrodes with a 250-Hz sampling rate and

band-pass filtered between 0.5 and 100 Hz from 9 subjects. The timing scheme consists of a fixation-cross 2 seconds followed by a period of MI task of 4s, which gives a cue for 1.25 seconds. The data set consists of two sessions of data, and each session has 288 trials (72 trials per class) for training and evaluation data.

4.2 SVM Classifier optimized by PSO-SA

K-fold cross-validation is introduced to ensure the universality of the algorithm. Namely, the test set and the training set are mixed together and then divided into K equal parts, 80% of which are randomly taken as the training set and the remaining 20% of which are used as the test set. Repeat the above steps of 5-fold cross-validation five times to avoid the effect of special cases on the experiment.

Assuming that the training set is $(x_1, y_1), \dots, (x_n, y_n)$ and the classifier has M kernel functions, where $x_i \in R^n$ is multi-domain feature fusion vectors of EEG signals in the input space X, and $y_i \in \{1, \dots, m\}$ is the target value of m classes.

Pair classification is a multi-classification SVM algorithm based on binary SVM, which is adopted by LibSVM library[15]. This one-against-one method constructs a binary SVM between each of the two classes. This method constructs $m(m-1)/2$ classifiers[16].

The multiple kernels make the feature vectors of the time-frequency and spatial domain fully reflected, solving the problem of heterogeneous information and unbalanced distribution in the MI data of EEG signals. The multiple kernel function is:

$$K(x_i, x) = \sum_{m=1}^M \beta_m K_m(x_i, x) \quad (8)$$

$$\beta_m \geq 0, \sum_{m=1}^M \beta_m = 1$$

where β_i is the fusion coefficient of the i^{th} kernel. In equation (8), the weighed linear synthetic kernel replaces the single kernel function, and different kernel functions and free parameters are selected for each basic kernel function.

In order to search the parameters accurately, SVM classifier is optimized by Particle Swarm Optimization (PSO) and Simulated Annealing (SA) algorithm[17] in this paper.

4.3 Result and Analysis

The experimental data set IIIa in BCI Composition 2005 are classified with four kind of feature fusion algorithms in

the Fig 5. The average accuracy of five times is the final accuracy. The final classification results are as follows:

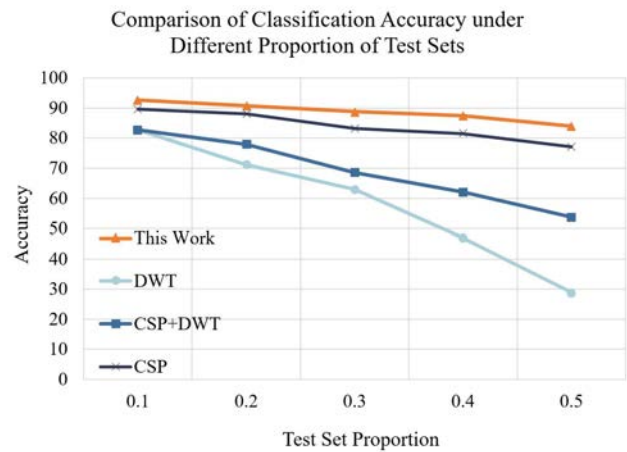


Fig. 5: Comparison of classification accuracy under different proportion of test sets in BCI Composition 2005

Fig.5 shows that the classification accuracy of various algorithms improves when the proportion of the test set decreases. Moreover, when the proportion of test set is 0.5, the classification accuracy of the proposed algorithm in this work is more than 80%, which is much higher than other algorithms. The accuracy of the algorithm in this paper is also higher than other algorithms in other proportion of test sets.

To test the performance of the algorithm with EEG signals of different subjects, the k3b and k6b EEG dataset of the subjects in BCI2005desc-IIIa and Graz dataset of the subject8 in BCI2008desc-IV 2a are used.

In the course of the experiment, the algorithm in this paper is compared with the other three feature fusion algorithms in Table 2. The control group consists of single algorithm, such as DWT or CSP, and CSP+DWT where the concat of feature vectors extracted by CSP and DWT which is called CSP+DWT. They are designed to compare the accuracy of SVM classification among the features of one single domain in the time-frequency or spatial domain and the fused features of both domains with different fusion methods.

The results illustrate that the classification performance of multi-domain feature fusion model based on CCA are better than other algorithms. The accuracy of single domain feature is less than 90%. The accuracy of fused feature is improved under different datasets in varied degree and better than any of single-domain feature or

Table 2: The performance compared on BCI Competition data set among algorithms

Algorithm		Accuracy			Average
Feature Extraction	Classifier	BCI Competition 2005		BCI Competition 2008	
		K3b	K6b	Graz	
CSP	MKL-SVM	88.14%	85.71%	61.43%	78.42%
DWT	MKL-SVM	71.19%	71.43%	72.14%	71.59%
CSP + DWT	MKL-SVM	77.97%	77.14%	70.36%	75.16%
This Work		93.22%	87.50%	78.07%	86.26%

feature extraction algorithm. This shows that fused features can be more effective in classifying MI EEG signals.

Table 3: The performance compared on BCI2008desc-IIa EEG data set among algorithms

Method	Accuracy
FBCSP[18]	67.75%
SVM+FBCSP[19]	71.78%
C2CM[19]	74.46%
NCSP[20]	67.01%
AX-LSTM[21]	76.47%
FDBN[22]	72.18%
3D CNN[23]	75.00%
Multi-CNN Feature Fusion[24]	74.80%
This Work	78.07%

Table 3 shows the comparison and verification results of the proposed algorithm and different algorithms under BCI Competition 2008-IIa. It can be seen that the proposed algorithm is not only superior to the 7 kinds of algorithms without feature fusion in the Table 3, but also superior to the Multi-CNN algorithm using feature fusion, reflecting the advantages of the feature fusion based on CCA algorithm proposed by this work.

5 Conclusion

In this paper, a feature fusion algorithm based on CCA is proposed for EEG MI classification. The experiment above demonstrates that CCA is able to extract some abstract representations of features. Compared with the single domain MI EEG signals, the fused features improve the overall classification accuracy when these extracted features are fused. The feature fusion algorithm proposed in this paper is better than other feature fusion models in MI EEG multi-classification for specific tasks. The accuracy of more than 87.50% is obtained on BCI2005desc-IIIa, and 78.07 on BCI2008desc-IIa. This algorithm can be used to analyze the common features of similar EEG signals, so the classification results of this system are significantly improved.

References

- [1] Mattia D, Astolfi L, Toppi J, et al. Interfacing brain and computer in neurorehabilitation[C]// 2016 4th International Winter Conference on Brain-Computer Interface (BCI). IEEE, 2016.
- [2] Chong Lee. Research on visual guidance and pattern classification of motor imagery EEG[D]. Chongqing University of Posts and Telecommunications, 2018.
- [3] Wanchang Deng. The Brain-Computer Interface research on Event related Potential[D]. Beijing Jiaotong University, 2009.
- [4] Pitsik E, Frolov N, Badarin A, et al. Network analysis of electrical activity in brain motor cortex during motor execution and motor imagery of elderly[C]// 2020 International Conference "Nonlinearity, Information and Robotics" (NIR). 2020.
- [5] Peng Zhou. The Research of Brain-Computer Interface Based on Motor Imagery[D]. School of Precision Instrument and Optoelectronics Engineering, Tianjin University, 2007.
- [6] Xuetian Xu, Yuesi Cai. Motor imagery recognition based on graph convolution network[J]. Computer Engineering and Applications, 2021, 1:1-8. Jasmin Kevric, Abdulhamit Subasi. Comparison of signal decomposition methods in classification of EEG signals for motor-imagery BCI system. 2017, 31:398-406.
- [7] Xinghan Shao. Research on Signal Processing Methods of Brain-Computer Interface Based on SSVEP [D]. Shandong University, 2020.
- [8] Ming Meng, Qingjun Zhu, Qingshan She, etc. Two-level Feature Extraction Method for Multi-class Motor Imagery EEG[J]. ACTA AUTOMATICA SINICA, 2016, 12: 1915-1922.
- [9] Wang K, Zhai D H, Xia Y. Motor Imagination EEG Recognition Algorithm based on DWT, CSP and Extreme Learning Machine[C]// 2019 Chinese Control Conference (CCC). IEEE, 2019.
- [10] Graimann B, Allison B, Pfurtscheller G. Brain-Computer Interfaces: Revolutionizing Human-Computer Interaction[M]. Springer Publishing Company, Incorporated, 2013.
- [11] Xianfa Zhang, Kuangrong He, Lei Chen. Motor Imagery EEG Classification Based on Immune Multi-domain-feature Fusion and Multiple Kernel Learning SVM[J]. ACTA AUTOMATICA SINICA, 2020, v.46(11):181-190.
- [12] Chetana Kamlaskar, Aditya Abhyankar, et al. Multimodal System Framework for Feature Level Fusion based on CCA with SVM Classifier[J]. IEEE-HYDCON, 2020, 10:1-8.
- [13] Benjamin Blankertz. BCI Competition III[EB/OL]. http://www.bbc.de/competition/iii/#data_set_iiib, 2005.
- [14] Benjamin Blankertz. BCI Competition IV [EB/OL]. <http://www.bbc.de/competition/iv/>, 2008.
- [15] Chang C C C, Lin C C C. LIBSVM: A library for support vector machines[J]. 2011.
- [16] Chih-Wei, Hsu, Chih-Jen, et al. A Comparison of Methods for Multiclass Support Vector Machines. [J]. IEEE Transactions on Neural Networks, 2002.
- [17] Gelatt, M. P. Vecchi, S. Kirkpatrick, C. D. Optimization by Simulated Annealing[J]. Science, 1983, 220(4598):671-680.
- [18] Park S H, Lee D, Lee S G. Filter Bank Regularized Common Spatial Pattern Ensemble for Small Sample Motor Imagery Classification[J]. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2018, 26(2):498-505.
- [19] Siavash, Sakhavi, Cuntai, et al. Learning Temporal Information for Brain-Computer Interface Using Convolutional Neural Networks. [J]. IEEE Transactions on Neural Networks & Learning Systems, 2018.
- [20] Yang H, Sakhavi S, Ang K K, et al. On the use of convolutional neural networks and augmented CSP features for multi-class motor imagery of EEG signals classification[C]// 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society IEEE EMBS. IEEE, 2015.
- [21] Wang P, Jiang A, Liu X, et al. LSTM-Based EEG Classification in Motor Imagery Tasks[J]. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2018, PP (11):1-1.
- [22] Lu N, Li T, Ren X, et al. A Deep Learning Scheme for Motor Imagery Classification based on Restricted Boltzmann Machines[J]. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2016.

- [23] Zhao X, Zhang H, Zhu G, et al. A Multi-Branch 3D Convolutional Neural Network for EEG-Based Motor Imagery Classification[J]. IEEE transactions on neural systems and rehabilitation engineering, 2019, 27(10):2164-2177.
- [24] Amin S U, Muhammad G, Abdul W, et al. Multi-CNN Feature Fusion for Efficient EEG Classification[C]// 2020 IEEE International Conference on Multimedia & Expo Workshops (ICMEW). IEEE, 2020.